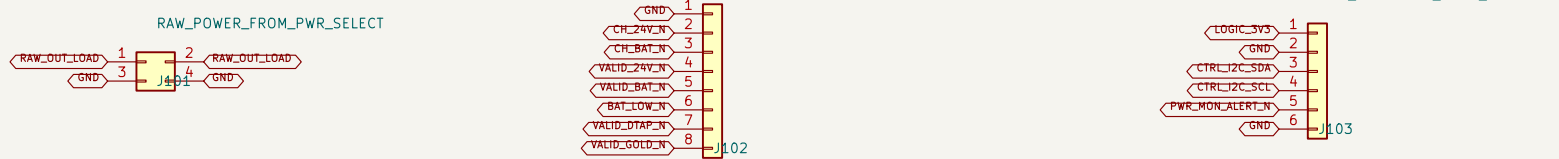


A. PWR-SELECT INTERFACE

D. DETAILED CAPTURE SUITE – REV A1

E. CONNECTED CM5-CARRIER HIERARCHY



Mates with PWR-SELECT (301) two contacts per priority, realized 10 A drawing.

Mates with PWR-SELECT (401) add 3.3 V pull-ups on the carrier.

Mates with PWR-SELECT (402) carrier supplies 3.3 V and even I2C pull-ups.

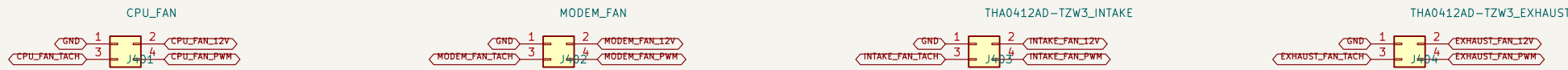
B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL



Mates with AUDIO-B08 (351) using Nixie #7832-6433 headers; harness assembly remains vibration-qualified.

Carrier drives HCU/RCU/FSMC/SEC data; AUDIO-B08 drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS



CPU_FAN

M02EM_FAN

TH04L1240-TZW3_INTAKE

TH04L1240-TZW3_EXHAUST

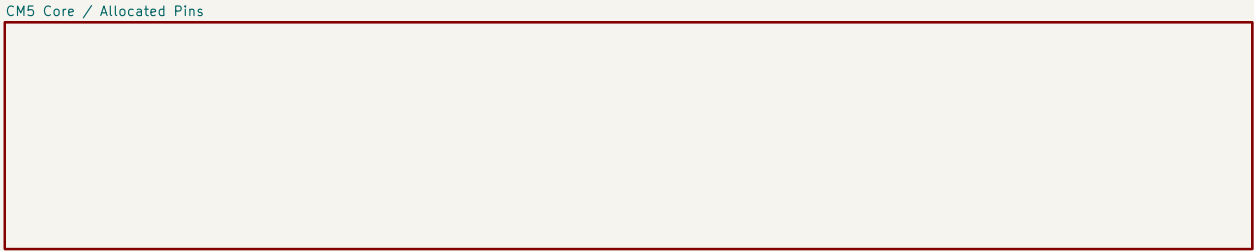
IN33 is physically keyed/tabbed INTAKE; IN46 is keyed/tabbed EXHAUST.

- 01 CM5-Carrier-Allocated exact 301-contact CM5 symbol, 76 named contacts (74 connected, 2 assigned NC)
- 02 CM5-Carrier-Allocated: VCC_SYSN, reset, recovery, power-on and debug UART
- 03 Network-PCie: 87C92G6080P, 3x LAN7430 and native WM6 front end
- 04 Network-PCie: 8x Marlin Magjack and Nixie 0879010102 Mini PCie 4xM Wi-Fi
- 05 WWAN-SIM: TE 219W20-13 8-key, USB3/USB2, controls and dual-SIM mux
- 06 Display-Harness HDMI, USB Touch and 4x4x4x4 1.2 V / 2.0 A branch
- 07 Audio-Control: I2S0 TDM, I2S1 ES8368 headset and CTA amplifier path
- 08 Thermal-ID: I2C translation, GPD expansion, sensors and four fan channels
- 09 Power-Regulators-AI: protected raw hold-up, all main rails, sequencing and test points
- 10 Audio-B08: all XLR channels bonded to chassis with one controlled AGND bond
- 11 Reserve: all three board tests have zero EDC violations; child control is allowed

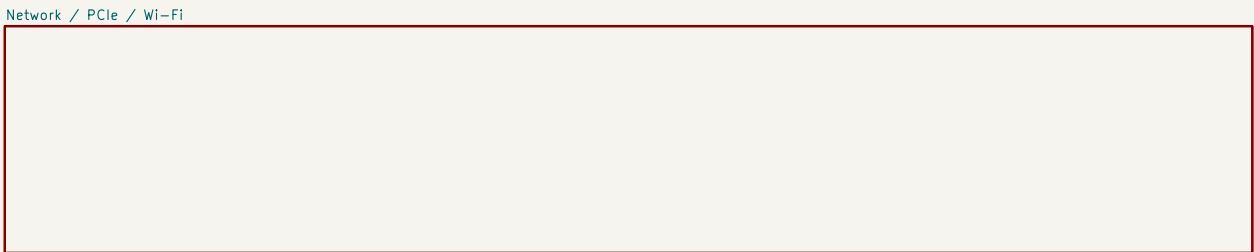
DC/DC compensation and magnetic remote a separate power-design calculation milestone.

High-speed routing starts only after startup and connector 2 domains are released.

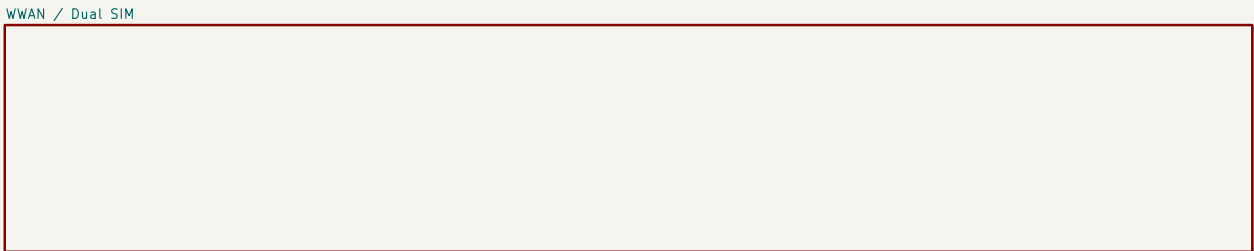
The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.



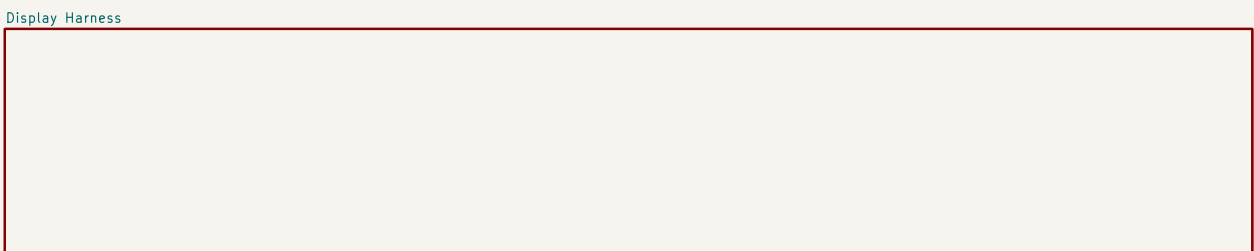
File: CMS-Carrier-Allocated.kicad_sch



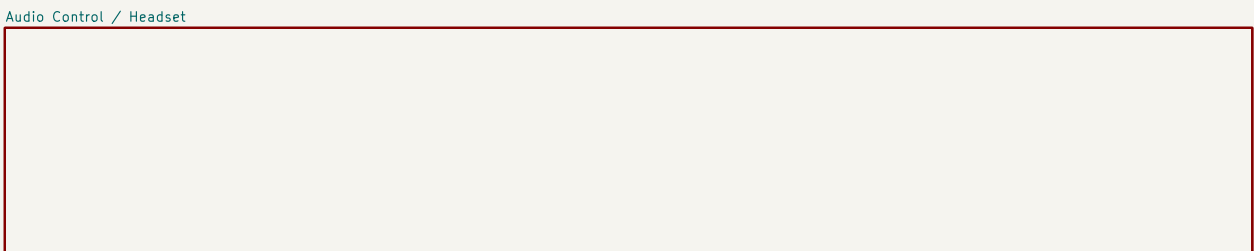
File: Network-PCie.kicad_sch



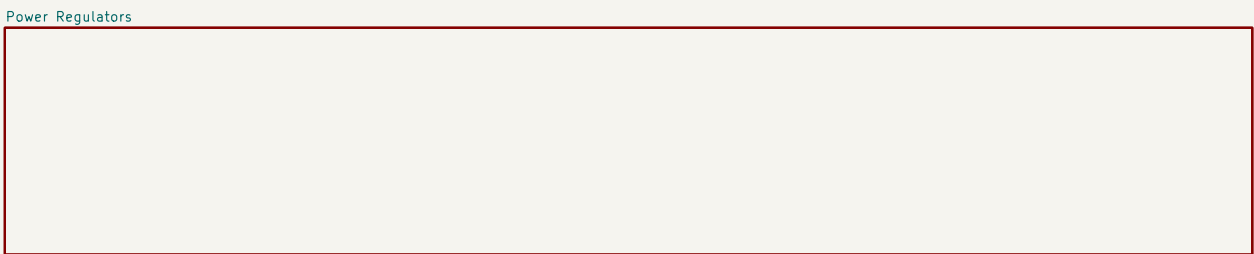
File: WWAN-SIM.kicad_sch



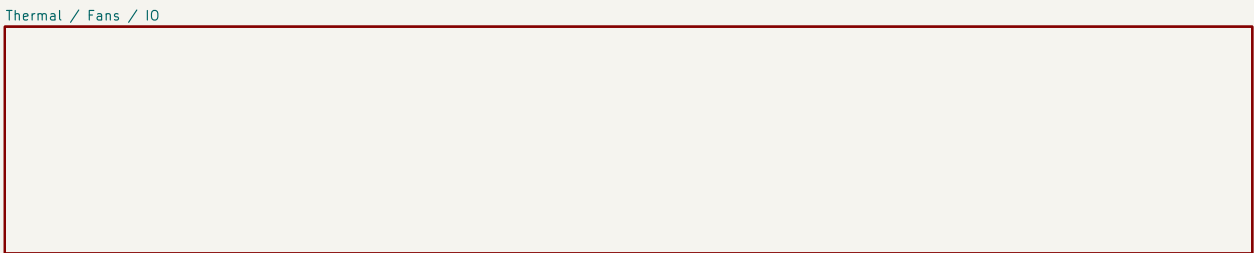
File: Display-Harness.kicad_sch



File: Audio-Control.kicad_sch



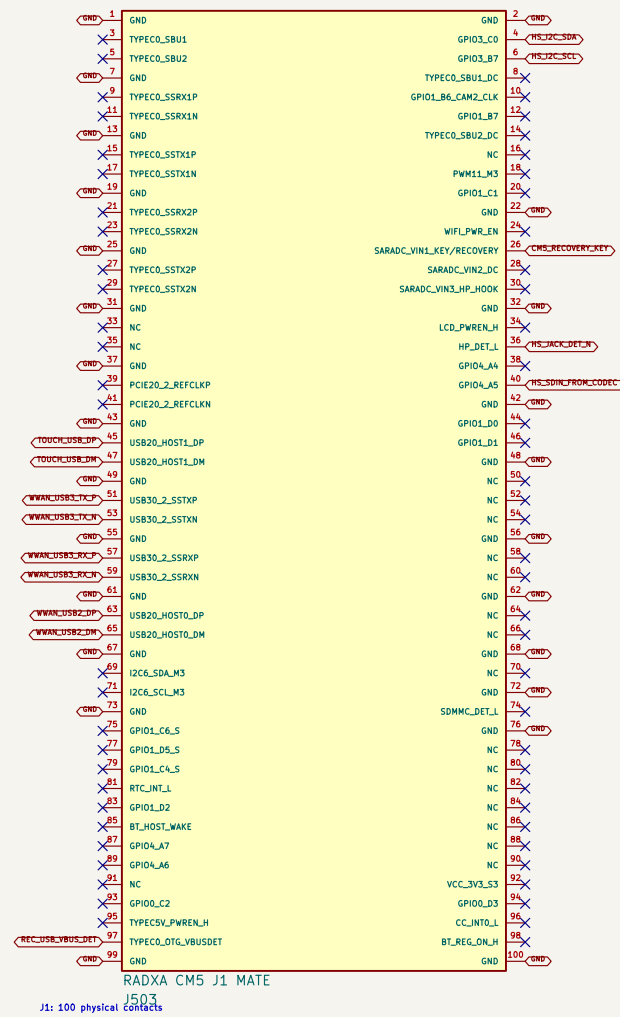
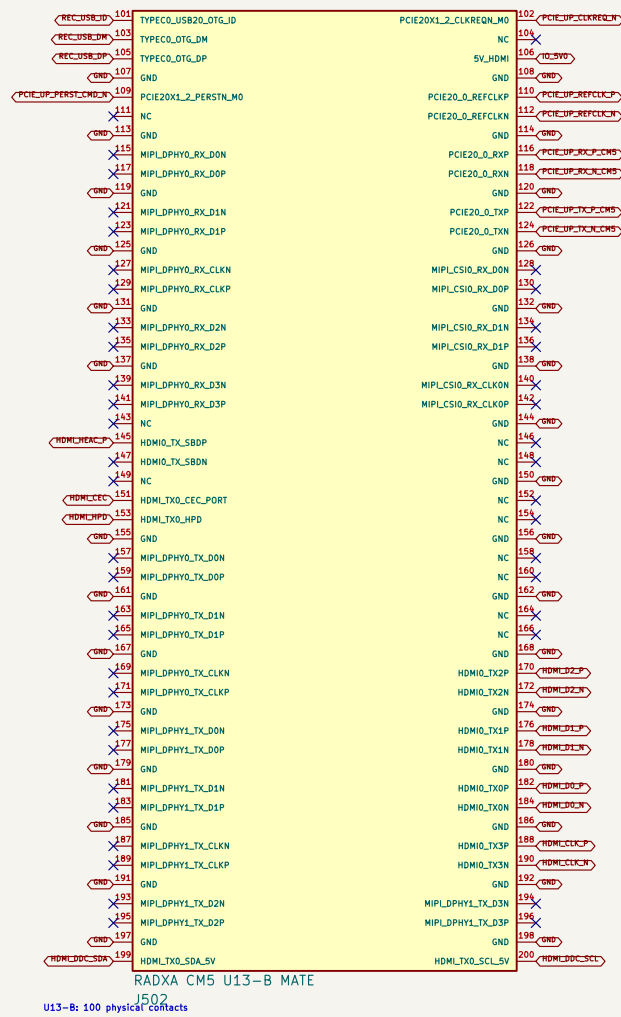
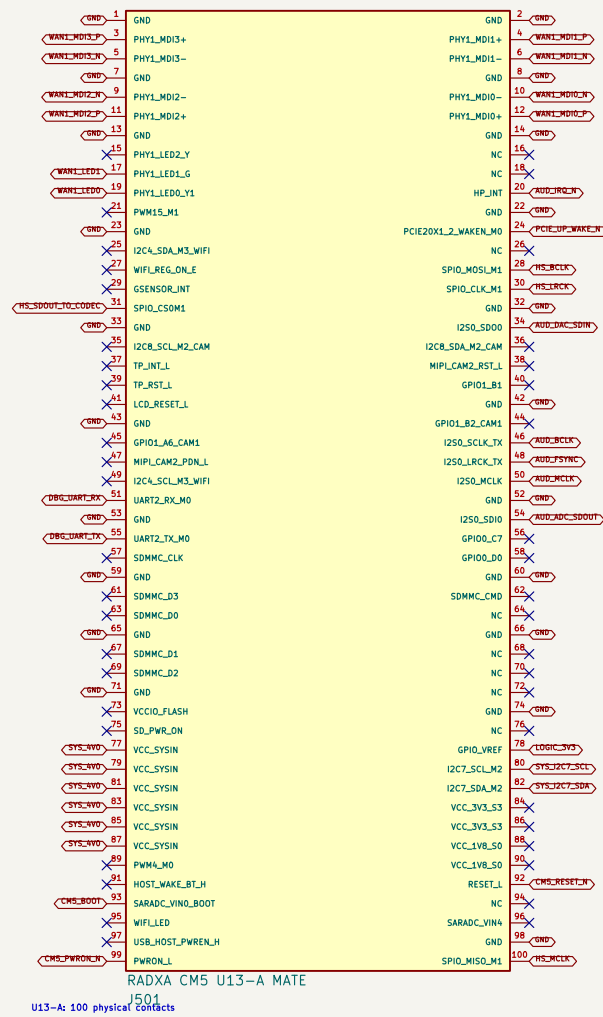
File: Power-Regulators-AI.kicad_sch



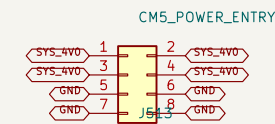
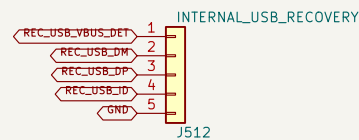
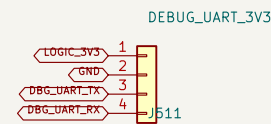
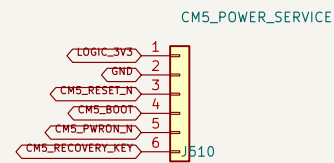
File: Thermal-ID.kicad_sch

USE/DOE/MS-1046 are Interface-only Application; the detailed-page connections over the board footprint.

1. EXACT RADXA CM5 MATING CONNECTORS



2. INTERNAL POWER / STARTUP / RECOVERY SERVICE



3.3 V UART only; keyed factory/debug cable

Internal keyed recovery harness; service fixture owns the protected USB-C receptacle and CC network.

Kelvin=sense SYS_4V0 at J501; 4.0 V follows the Radxa CM5 carrier design note.

3. OWNERSHIP AUDIT

76 / 76 contacts owned; 74 connected, 2 assigned NC, 0 duplicate physical-pin claims.

WAN1 uses native MDI. PCIe Gen2 x1 feeds the packet switch.

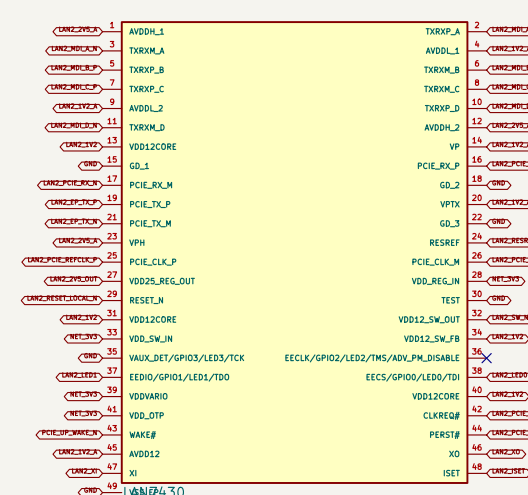
WWAN owns USB30_2 + USB20_HOST0; touch owns USB20_HOST1.

HDMI_TX0, I2S0 TDM, I2S1 headset, I2C7, I2C3 and UART2 are reserved exactly once.

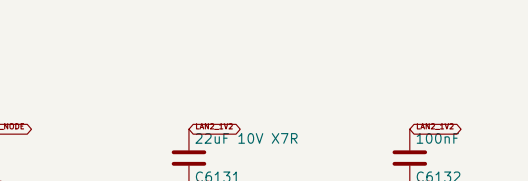
Pin 145 reaches HDMI pin 14; pin 147 is assigned NC because HEAC/ARC is unused. U13-B pin 106 receives IO_5V0.

Unused alternate-function contacts carry explicit no-connect markers on this A1 baseline

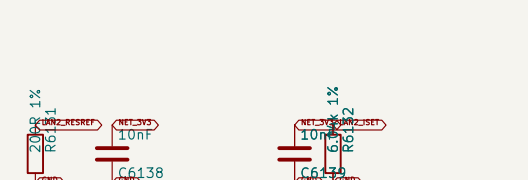
DETAILED CAPTURE BASELINE – verify CM5 voltage limits before release
All connector grounds join the carrier ground plane; allocated GPIO domain is 3.3 V
76 contacts are owned; 74 connected and 2 assigned no-connection; all other functions are
Three 100–contact Hirose DF40C mates; all physical pin numbers follow CM5 V2.21
ProComm
Sheet: /CM5 Core / Allocated Pins/
File: CM5–Core–Allocated.kicad_sch
Title: Radxa CM5 ProComm Carrier – Exact CM5 Connector Allocation
Size: A2 Date: 2026–08–13 Rev: A1
KiCad E.D.A. 10.0.5-rc1 Id: 2/8



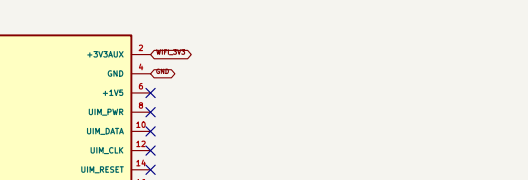
709011462 collap=mode 1, Gd: starts direct to char's, P0: side residual (1)



U660A: exact PCIe lanes, reset and distributed reference clock



mode strap, I2C and calibration; other straps use documented internal default:



GP0[1:0] = 01 selects 606 mode. GP01 uses its internal pulldown; GP00 is pulled high.

6601 rail bypass: one 100 nF per supply rail plus 22 uF bulk on 1.0 V and 3.3 V

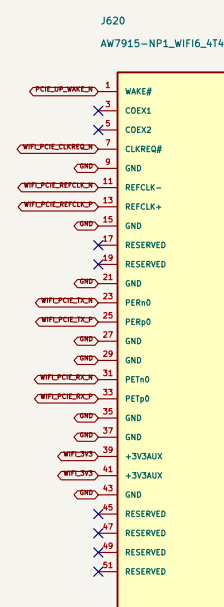
U008 asserts switch *PERST#* unless the CR5 request and both PCIe rail power-good signals are high.

CMS Emuware/100d-complex timing must keep the request low until REPCLK and nulls have been visible for at least 500 ms

PWR_FLAG symbols declare off-sheet regulator outputs for DRG; they are not physical parts.

PWR_FLAG

2620 uses the controlled SO-57910-001 C2 (and pattern); SO/First-article FR remains a release per

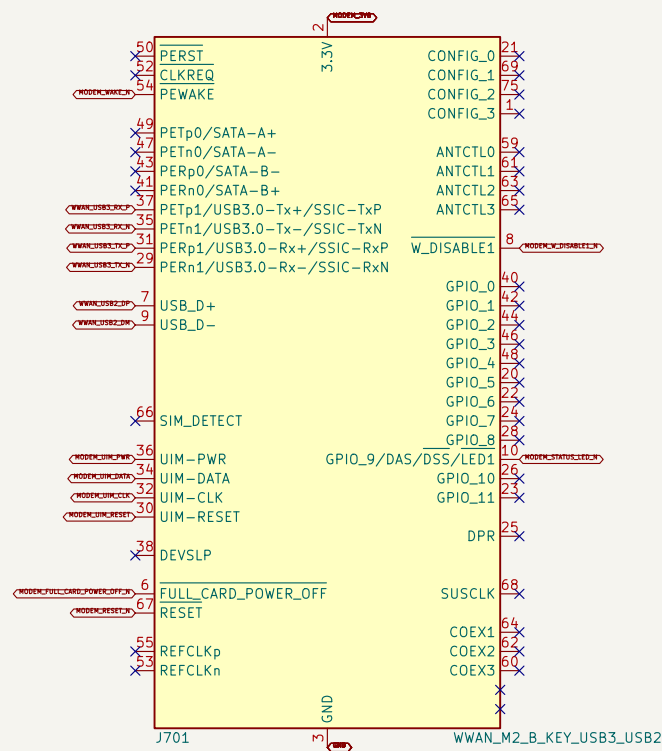


Port 5 is disabled and unrouted; generic cable headers are prohibited on PCIe Gen2 lanes.
Exact package balls/pins from Diodes DSA0210 Rev 8 and Microchip D500002631G
Native WAN1 plus three LAN7430 endpoints use exact W74991114412 MagJacks
P17C9X2G608GP in 606 mode: x1 upstream plus five x1 downstream ports

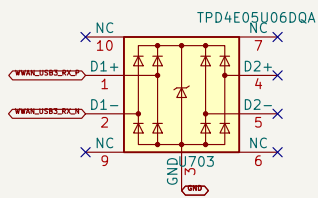
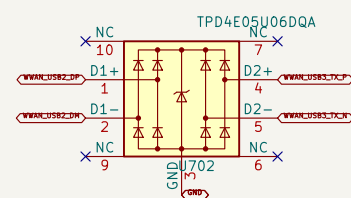
Sheet: /Network / PCIe / Wi-Fi/
File: Network-PCIe.kicad_sch

Title: Radxa CM5 ProComm Carrier - PCIe / Ethernet / Wi-Fi		
Size: A1	Date: 2026-08-14	Rev: A1
KiCad E.D.A. 10.0.5-rc1		Id: 3/8

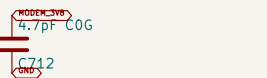
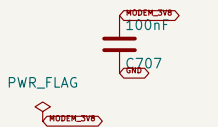
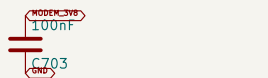
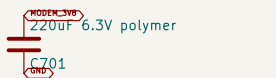
1. UNIVERSAL M.2 B–KEY WWAN SOCKET



2. USB ESD AND MODEM BULK CAPACITANCE



Place USB ESD at the modem socket: route USB 2 pairs as short, capacitance 90 pF into differential channels.



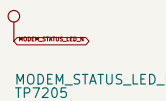
ESD: TPDAE05U060GA RESET/CLK0/CLK1 and channel VCC inside the modem shell bonds to CMASISLANS.



CMASISLANS and ESD represent the RDSM-GL, low-loss VCC reference network directly at R32 pins 2/4 and 7/9/1/3.

Power-Regulator CS08T CS08T refers CS08 of upstream bank. They do not replace the socket-mount 2 x 208 of unit RF module.

5. RF BULKHEAD HARNESS CONTRACT

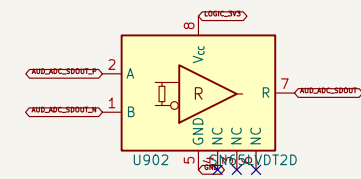
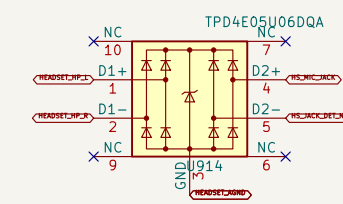
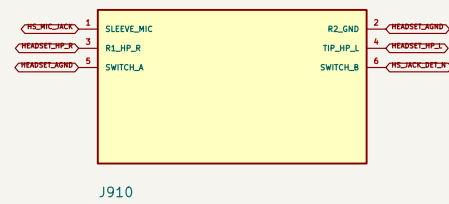
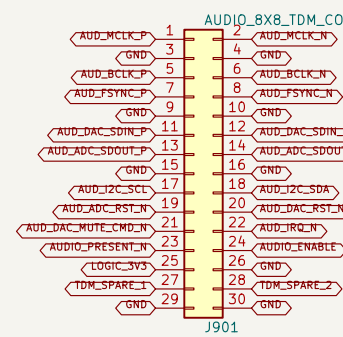
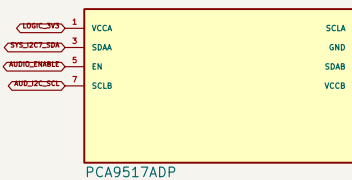
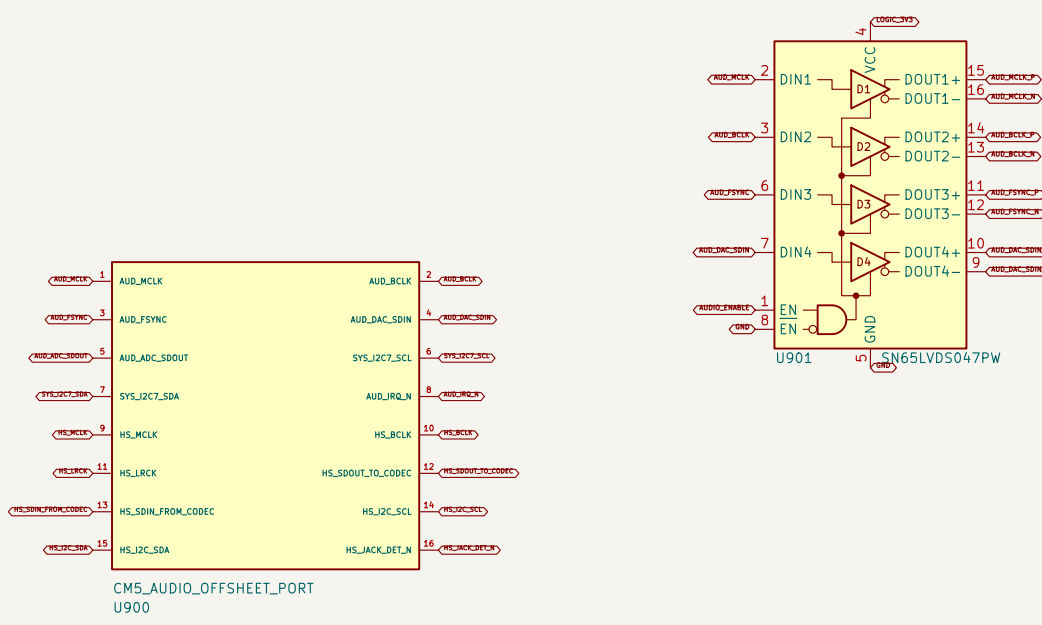


Four left-board EXT RDSM320A uses RF 10-to-10M bulkhead pluglets. Their cable length measures a needed sample gain.

RF coax leaves the module directly for four right-side bulkhead antennas
FSA2567 defaults to physical SIM 1; open-drain low selects physical SIM 2
Dedicated 3.0 V / 6 A rail; verify selected modem peak-current waveform
M.2 B-key modem socket supports USB 3.0 and USB 2.0 fallback

ProComm
Sheet: /ANWAN / Dual SIM/
File: WWAN-SIM.kicad_sch
Title: Radxa CM5 ProComm Carrier – WWAN and Dual SIM
Size: A1 Date: 2026-08-14 Rev: A1
Kicad E.D.A. 10.0.5-rc1 ID: 4/8

1. I250/TDM PROGRAM-AUDIO LVDS LINK



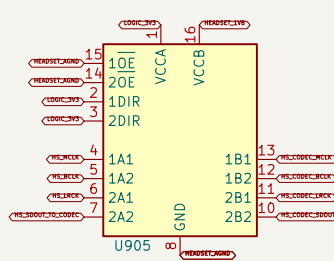
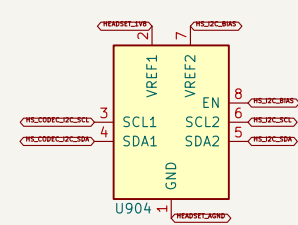
100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

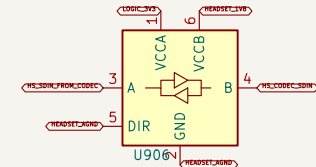
6. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD



2. I251 HEADSET LEVEL TRANSLATION

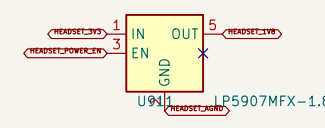
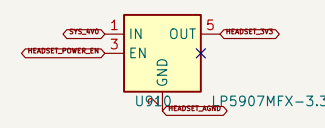


100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.



100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

3. DEDICATED LOW-NOISE HEADSET RAILS



100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.



7. HEADSET CONTROL POLICY

100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

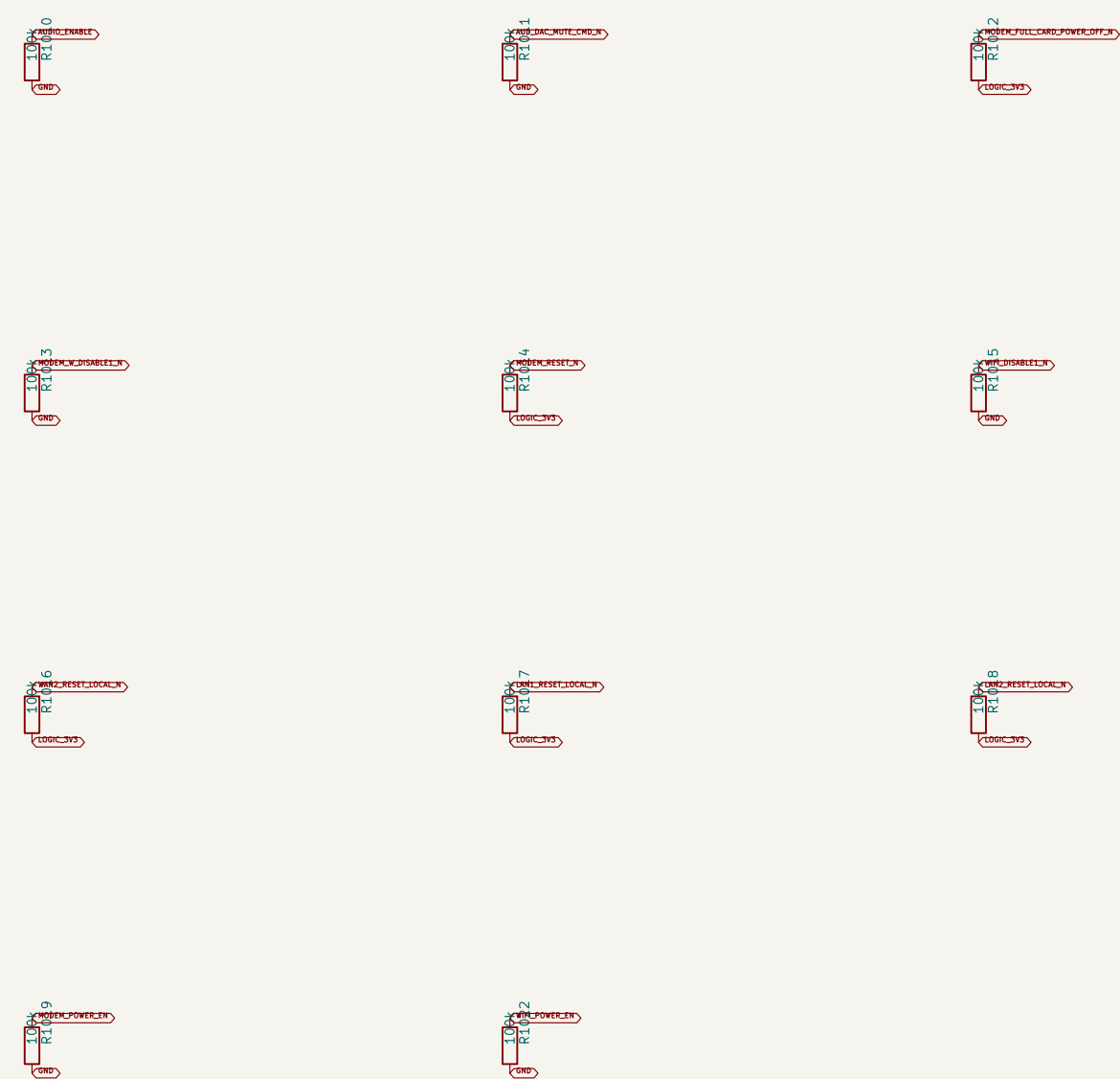
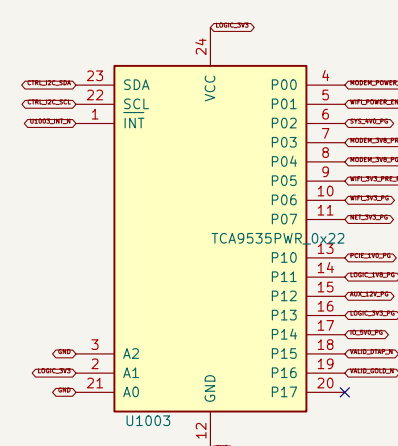
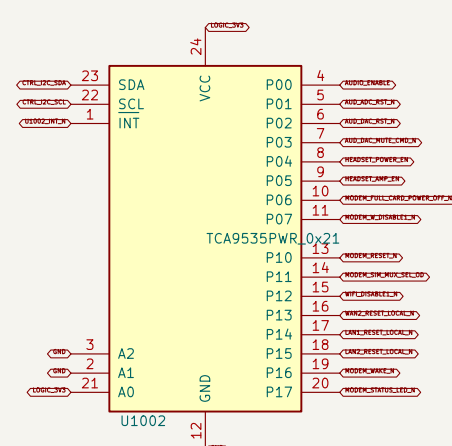
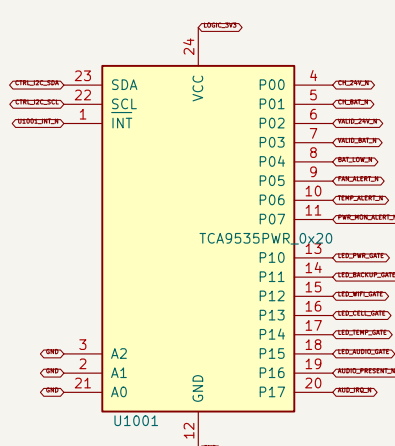
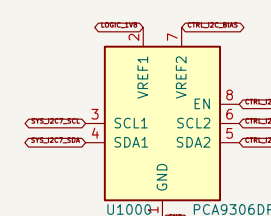


100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

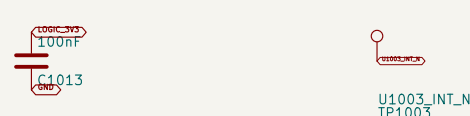
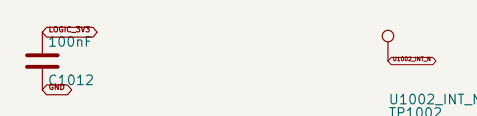
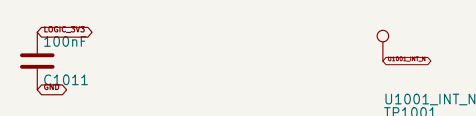
100% audio and video signal quality. Note: For 100% audio and video signal quality, the audio and video signals must be sent to the audio and video outputs.

1. 1.8 V TO 3.3 V SYSTEM I2C CONTROL BUS

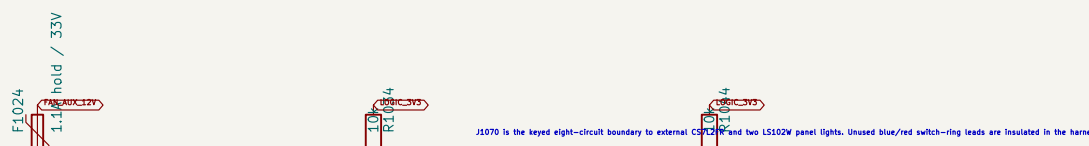
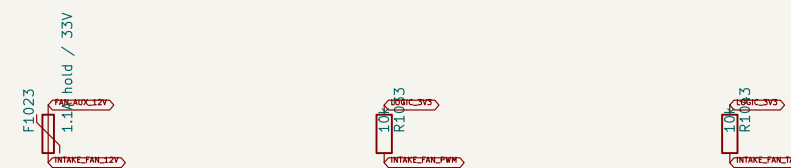
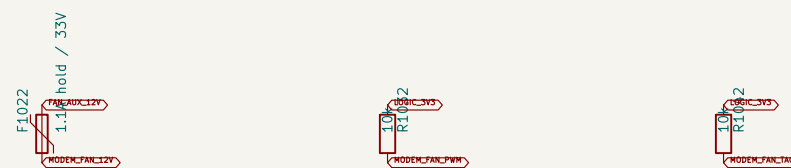
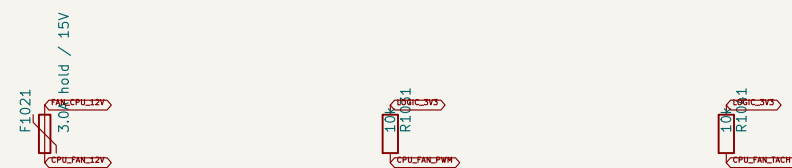
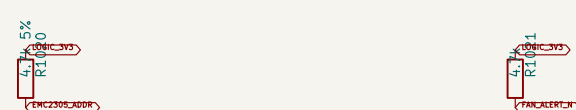
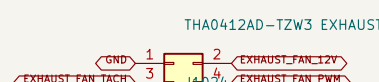
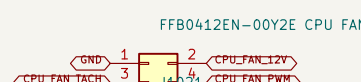
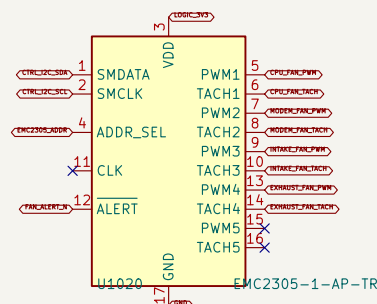
2. THREE 16-BIT GPIO EXPANDERS



The control branch is isolated to its own PDB550, which is not in contact with the separate 4000-800 branch prepared



3. EMC2305 FOUR-FAN CONTROL WITH FAILSAFE



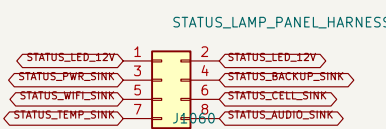
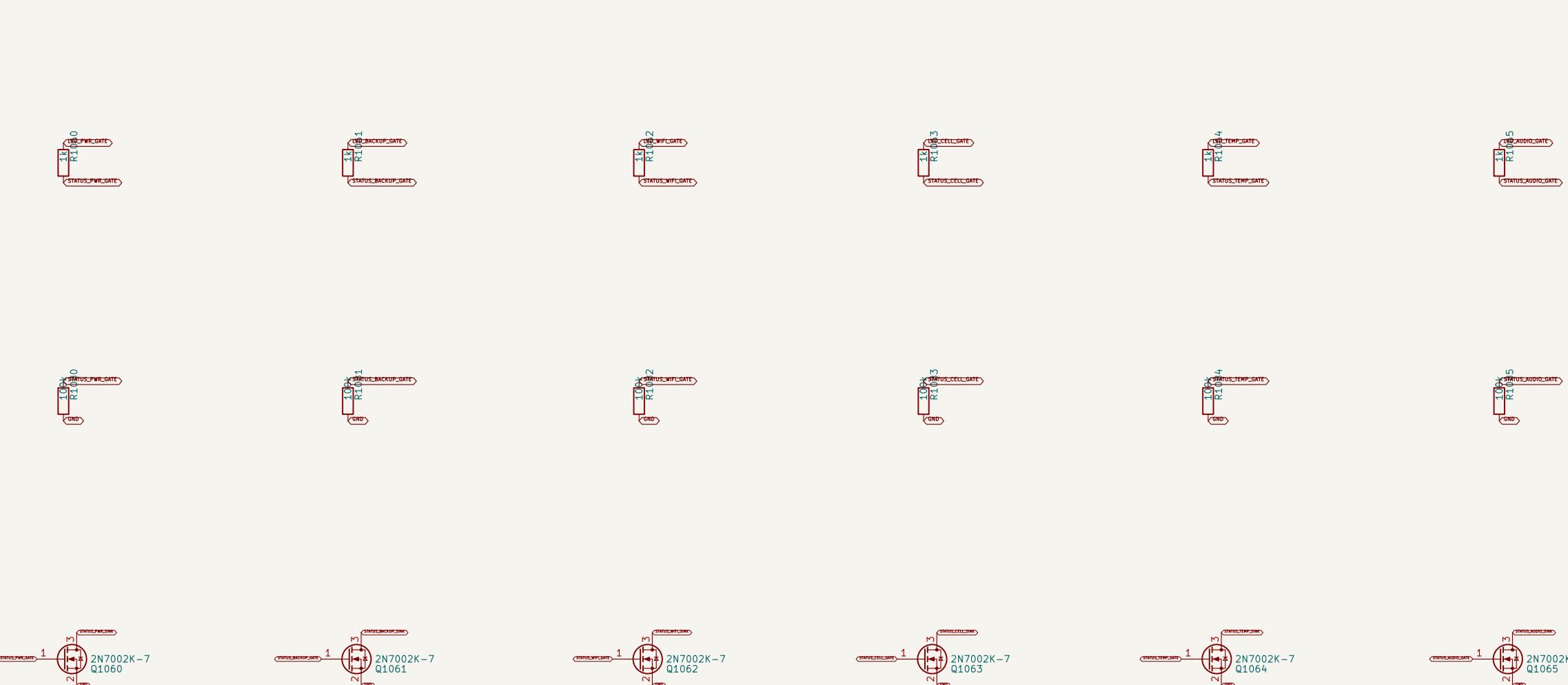
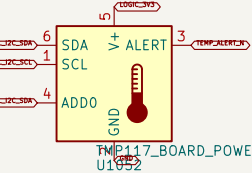
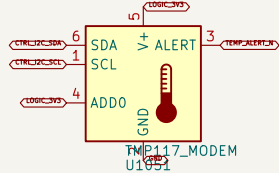
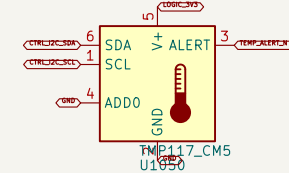
4004.03, 4.7% to 5.3 % (various 7-gh DMSu address 0-24). Internal clock is zero.

Exclusion criteria: 2430 site base / dollar 26 = 1304 site PPM direct only site test monitoring, 200K start and 20K minimum running only

Fun 1 is panel intake and fun 2 is panel exhaust. CPU and modem fans circulate locally beneath their dedicated mesh/foampad an

All four P&H units pull high when US300 is absent or uncovered, commanding the selected 4-wire fans to full speed.

4. THREE HIGH-ACCURACY TEMPERATURE SENSORS



TC49535 devices at 0x20/0x21/0x22 capture status and drive safe control outputs
TMP117 sensors monitor CMS, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware
EMC2305 controls CPU, modem, intake, and exhaust fans independently

ProComm	
Sheet: /Thermal / Fans / IO/ File: Thermal-IO.kicad_sch	
Title: Radxa CM5 ProComm Carrier – Thermal and GPIO Cont	
Size: A0	Date: 2026-08-14
KiCad E.D.A. 10.0.5-rc1	Rev: A1 Id: 8/8